Duration
0.20
Fragmentation (reversed)
0.15
Awakenings (reversed)
0.10
Fall asleep difficulty (reversed)
0.10
Wake clarity
0.10
Get out of bed difficulty (reversed)
0.10
Regularity
0.15
Environment
0.10

For each day t, define a set:
Vt={all variables that were actually recorded on day t}

Complete record date
Vt={D,F,A,I,C,G,R,E}

Partial record of days (forgot to fill in nighttime awakenings / bedtime)
Vt={D,F,C,G}

N(x)=x-min(x)/max(x)-min(x)

S(x)=1-N(x)
S(F)=1-N(F)
S(A)=1-N(A)
S(I)=1-N(I)
S(G)=1-N(G)
N(D)=N(D)
N(C)=N(C)
R,E

For a given day t,
retain only the variables in Vt
and perform weight normalization:

[redacted]

Missing variables are not included in the calculation for that day.
The weights of the remaining variables are automatically renormalized.
The index still reflects the "perceived and recorded physical condition" of the day.

NightBodyIndex=100×(0.20N(D)+0.15S(F)+0.10S(A)+0.10S(I)+0.10N(C)+0.10S(G)+0.15R+0.10E)

D = Sleep duration (hours, larger=better)
F = Sleep fragmentation / continuity (Q3, 0–4, larger=worse)
A = Night awakenings count (Wake Up at Night Q2, 0–4+, larger=worse)
I = Difficulty falling asleep (0–4, larger=worse)
C = Wake-up clarity (0–4, larger=better)
G = Difficulty getting out of bed (0–4, larger=worse)
R = Regularity (0–1, larger=better; calculated based on the deviation of bedtime + wake time)
E = Environment score (0–1, larger=better; heuristic mapping inspired by recored keywords)

Vividness: Q8 (0–4 → 0–1)
Intensity: Q9 (0–4 → 0–1)
Motif Complexity: Q7 (Number of keywords 1–5 → 0–1)
Affect Arousal: Q10 (Use a simple mapping: anxiety/fear high, calm low... → 0–1)

vivid 0.35
intensity 0.35
motif 0.15
arousal 0.15

wake_clarity: 0.45
get_out_bed_ease: 0.35
wake_reg: 0.20

A) wake_clarity
wake_clarity_norm = wake_clarity / 3
B) get_out_bed_ease
ease = 1 - difficulty/4
C) wake_reg
A 4-hour tolerance period (cap=240min) :
wake_reg = 1 - clip(|wake_time - median_wake|, 0, 240) / 240

mental clarity = original value / 4
sleepiness = 1 - original value / 4
creativity / focus / social desire / emotional stability = original value / 4

Mental clarity	0.20
Focus	0.20
Sleepiness	0.15
Creativity	0.15
Social desire	0.15
Emotional stability	0.15

Baseline: 25%
Intake behavior: 25%
Digestion response: 50%

Step 1: Normalization (0–1 scale)
All variables were normalized to a 0–1 range, where higher values indicate a more comfortable or less stressful bodily state.
For 0–4 rating scales:
Symptom-based variables:
score = 1-(value / 4)
Comfort or positive variables:
score = value / 4
Yes/No variables (e.g. reflux, screen usage):
No = 1, Yes = 0
Meal duration:
An optimal duration of 20 minutes was assumed. Scores decrease as duration deviates from this value.
Fullness:
A moderate fullness level was treated as optimal. Deviation in either direction reduced the score.

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1.29 1.30 1.31 2.1 2.2 2.3 2.5 2.6 2.7 2.8 2.9